Atharva Wani

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Education

Master of Science in Robotics and Autonomous Systems: Mechanical and Aerospace Engineering — Expected May 2024
Arizona State University, Tempe, AZ — 4.0 GPA
Ira A. Fulton College of Engineering
Relevant Coursework: *Programming of IoT devices, Wearable devices for health, Mechatronic device innovation, Expressive robotics*, Multi-Robot Systems, Advanced System design and modelling, Machine learning and Deep learning.

Bachelor of Technology Mechanical Engineering: Machine Design minor — Graduated August 2021
Manipal Academy for Higher Education, Manipal, Karnataka, India. — 3.45 GPA
Manipal Institute of Technology
Relevant Coursework: DFMEA, Tribology, Fatigue and Fracture, Design of mechanical systems, introduction to mechatronics, Artificial intelligence

Technical Skills

Design and modelling tools: SolidWorks, Catia, Creo, AutoCAD.
Prototyping tools: Arduino platform, ESP-32.
Design Analysis and simulation tools: SolidWorks, Ansys, Fusion360, Gazebo.
Programming languages: Python (Coursera certification), ROS2 (Udemy), JavaScript (NodeJS), MATLAB, C++, Java.
Financial and management courses: Essentials of Management, Engineering economics and financial management.
Research: Connected Automated Vehicles.

Experience

Tech trainee intern, Tesla Inc. — August 2023 – present

- Closely understanding the design concepts of electrified powertrains and system interactions in electric vehicles.
- Conducting comprehensive software and hardware diagnostics and implementing targeted solutions for Tesla vehicles to deliver swift, effective and customer-centric service.

Teaching Aide and Grader, EGR 316 and EGR 463 — Jan 2023 – August 2023

Engineering research intern, ACME process systems pvt. Ltd. — Feb 2021 – Aug 2021

Led the new R&D department with a small team of about 5 people.

- Analysed the performance of existing designs and develop new designs to compare them and build a database of all results. **(AutoCAD, SolidWorks, Ansys-Fluent)**
- Modelled and analysed over 10 impeller designs and helped team members devise a workflow for future work, laying the groundwork to fuel innovation over the coming years which has the potential to increase revenue by over 50% within a year of launching new models.

Industrial trainee, Precision Automation and Robotics India, pvt. ltd. (PARI Robotics) — Jan 2021 – Feb 2021

- Intensely involved in the workings of 9 departments going over all phases of engineering from analysis and testing of the design to the quality control, sale, and deployment of the product.

Automotive technician, Lonkar Automobiles pvt. ltd. — Jun 2019 – Jul 2019

Apprentice as an automotive technician as a part of my undergraduate curriculum.

- Learnt to fix engines, gearboxes, air conditioning systems and other preventive maintenance of cars.
- The highlight project of this apprenticeship is the reboring and rebuild of a hydro-locked Toyota diesel engine.

Vehicle dynamics engineer, Team Manipal racing — Aug 2018 – May 2019

Part of the Vehicle dynamics subsystem at the student racing team.

- Designed suspension geometry for the car and suspension components. The new design helped in cutting down up to 20% weight within the steering system with increased performance. **(WinGeo, SolidWorks)**
- Our car won 3rd place in national and international Baja-SAE competitions.

Projects

- **Product development of a neurological disability therapy device** in collaboration with Barrow neurological Institute, Phoenix, AZ. (ASU, AZ, USA) – Leading the team of students involved in full Product development of a home-use medical device for people affected by chronic weakness of hand due to stroke or spinal cord. Developed an innovative device using the e-stim technology solving the problems of wet e-stim pads. **(Arduino Nano, SolidWorks, Fusion360, 3-D printing (PLA, metal SLS), Laser cutting and etching.)**
- **Expressive robotics** (ASU, AZ, USA)**–** Designed robots and robotic systems that convey expressions. Worked on projects such as mechanical art pieces, human face muscle simulator robot, robots conveying body language. Involved in designing intricate systems to create precise movements. **(Arduino UNO, SolidWorks, Fusion360, 3-D printing, Laser cutting and etching.)**
- **Mobile robot with manipulator arm simulation. (ROS2, Gazebo, Moveit2)**
- **Wildlife camera trap with live stream on mobile app. (ESP-32, Arduino IDE, NodeJS, AWS, HTML, CSS)**
- **Thesis research topic –** Decision making by an autonomous vehicle in ethically challenging situation.
- Image compression, multi-robot communication, balancing an inverted pendulum using PID and LQR controllers (ASU, AZ, USA) **(ROS2, Gazebo, MATLAB)**
- **3D printing – Design optimization and Plastic and metal printing. Laser cutting and etching.** (ASU, AZ, USA)
- **Designed and built a Robo-wars robot for an inter-college competition during college tech-fest, Tech-Tatva.**